

École Pour l'Informatique et les Techniques Avancées – EPITA

Masters program

Course: Introduction to Network protocols and Architecture (INPA)

INPA Course Outline

Course schedule (tentative)

Date	No.	Topics	Duration (in hours)
(check Zeus platform for the date & time)	1	Primer, TCP/IP stack, TCP/IP stack operations, Class exercises	3
(check Zeus platform for the date & time)	2	IP address allocation & routing, Subnetting, ARP, Class exercises	3
(check Zeus platform for the date & time)	3	DHCP, Packet handling, Routing Protocols (RIP, OSPF), Class exercises	3
(check Zeus platform for the date & time)	4	TCP, UDP, ICMP, ON, Class exercise	3
(check Zeus platform for the date & time)	5	Internet layer protocols (IGP vs EGP, BGP), ACLs, Class exercises	3
(check Zeus platform for the date & time)	6	Application Layer protocols/Security (HTTP/S), Class exercise	3
(check Zeus platform for the date & time)	7	Application Layer protocols/Security (SMTP + PGP), DNS, Class exercises	3
(check Zeus platform for the date & time)	8	NAT, Class exercise, Review/Practice-session	3
Total			24 hours
(check Zeus platform for the date & time)	EXAM		1

GRADING criteria:

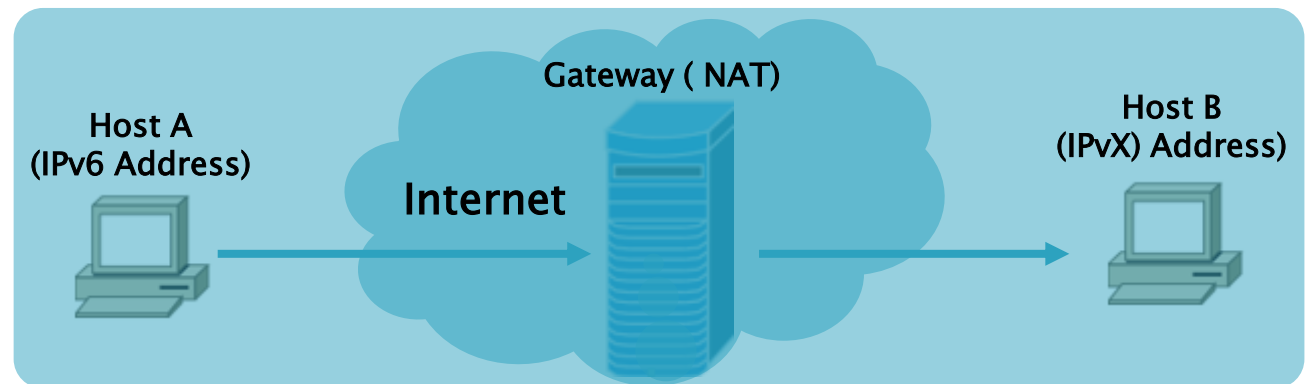
Class attendance and responsiveness:	10%
Class assignments (practical work):	40%
Final evaluation (Quiz & Exam):	50%

Lecture 8 Outline

- ▶ **Network Address Translation (NAT)**
 - Types
 - One-to-Many NAT
 - Class exercise 15
- ▶ Run-down
 - Open session

NAT (Network Address Translation)

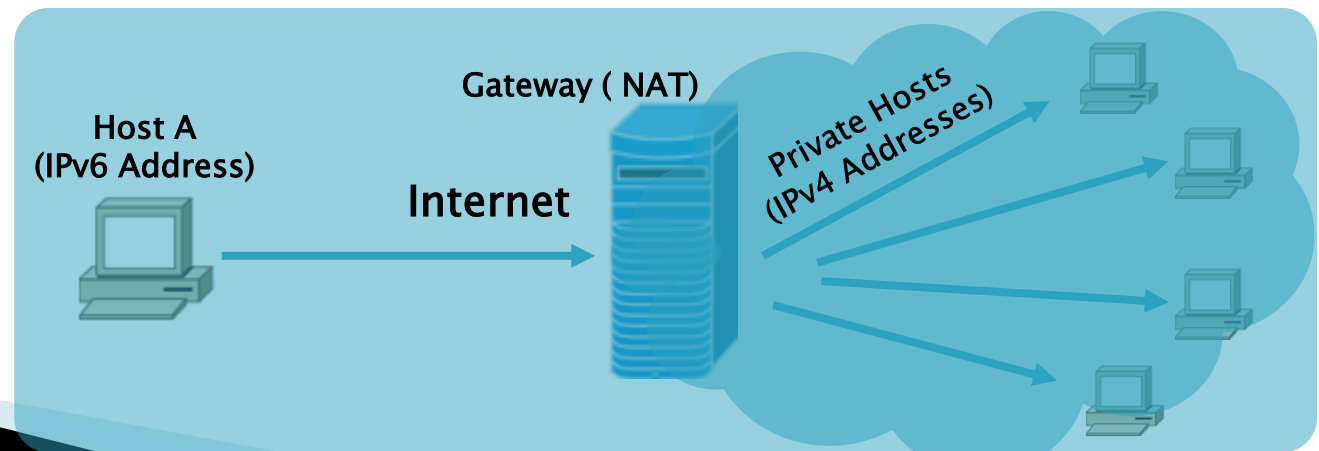
- ▶ A built-in component of most Routers
- ▶ NAT main Types:
 - **One-to-One NAT (a.k.a Basic or Static NAT):**
 - Enables interconnection of two incompatibly addressed (e.g., IPv4 to IPv6 & vice versa) or different assigned IP networks
 - Follows RFC 2663



NAT (cont.)

► NAT main Types:

- **One-to-Many NAT (A.k.a. IP Masquerading / Overload):**
 - IP address space (with many private IP addresses) hidden behind a single public IP address
 - Uses Port Address Translation (PAT): Inside Local and Inside Global PAT table (each local device is mapped with the public IP address using a unique port number)



One-to-Many NAT

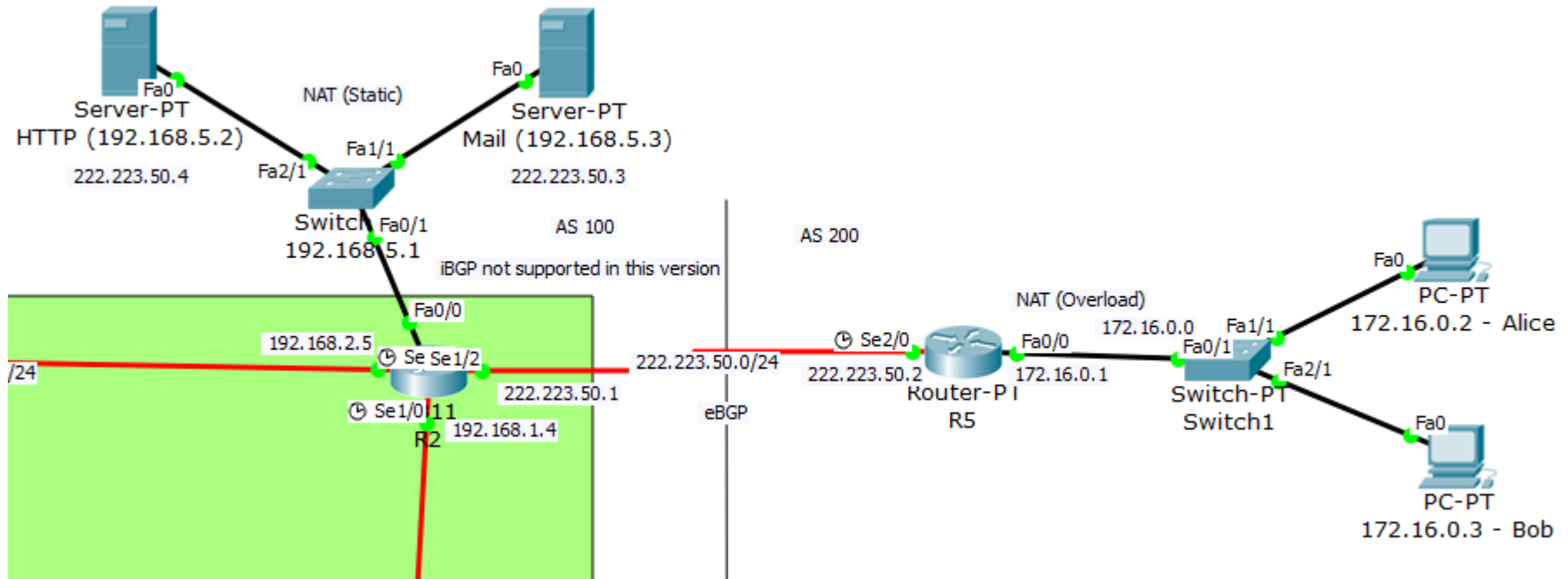
▶ Pro's:

- Provides enhanced security
- Facilitate the process of saving public IPv4 addresses
 - Partial solution for the IPv4 address space exhaustion problem
 - ...

▶ Con's

- Creates overhead/latency
- Difficult to manage with protocols that detect changes in headers (e.g., IPSec, ...)
- ...

Exercise 15: Practical work (1 / 2)



Inside Local: Fa0/0
Inside Global: Se1/2
Outside Local: Fa0/0
Outside Global: Se2/0

Exercise 15: Practical work (2/2)

► Using your last cisco packet tracer file:

1. Apply NAT overload in AS200 for NID: 172.16.0.0

```
R# (config-if)#ip nat inside //interface: inside local e.g., Fa0/0
R# (config-if)#ip nat outside //interface: inside global e.g., Se6/0
R# (config)# access-list NUM permit IP_OR_Net-ID Wild_Card
//Using access list to designate inside local IP address (or 'Interesting traffic ') to
be used for dynamic translation for NAT
R# (config)# ip NAT inside source list NUM interface INT_Num overload
//NATing using the access list for 'local inside' devices with the IP address of 'outside
local' interface IP
```

2. Apply NAT Static on HTTP and Mail server on AS100, and restrict these services for AS 200 only

```
R# (config-if)#ip nat inside //interface: inside local e.g., Fa0/0
R# (config-if)#ip nat outside //interface: inside global e.g., Se6/0
R# (config)# ip NAT inside source static Private_IP_Address
Public_IP_Address
R# Show ip nat translations
```

Verify from outside networks using PING or TRACERT/TRACEROUTE.

*Save cisco packet tracer
file with your 'First_Last-
name_Ex-3-15*

**Deadline: See
'Teams'
Assignment section**

i **Static NAT: We can initiate translation from both inside to outside and vice versa. That is not desired in Dynamic NAT and Overload (PAT)**

Lecture 8 Outline

- ▶ Network Address Translation (NAT)
 - Types
 - One-to-Many NAT
 - Class exercise 15
- ▶ **Run-down**
 - **Open session**

Open session

- Review
- Q/A

Lecture 8 ends here

- ▶ **Course Slides: Go to MS Teams:**
Introduction to Network protocols and architecture
→ Files section
- ▶ **Send your questions by email**
OR via direct message using MS Teams
- ▶ **Thank You!**

General reference list

- ▶ Internet Engineering Task Force. www.ietf.org.
- ▶ Wikipedia contributors. Wikipedia. Retrieved from <https://www.wikipedia.org/>
- ▶ University of Ottawa+2BCcampus Open Publishing+2Creative Commons+2
- ▶ IPv6.com. Retrieved from <http://ipv6.com>
- ▶ Cisco Systems. (n.d.). Resolve IP Fragmentation, MTU, MSS, and PMTUD Issues with GRE and IPSEC. Retrieved from <http://www.cisco.com/c/en/us/support/docs/ip/generic-routing-encapsulationgre/25885-pmtud-ipfrag.html> [Accessed June 1, 2015]
- ▶ Microsoft. (n.d.). IPv6 Technical Reference. Retrieved from [https://technet.microsoft.com/en-us/library/cc738109\(v=ws.10\).aspx](https://technet.microsoft.com/en-us/library/cc738109(v=ws.10).aspx) [Accessed June 1, 2015]
- ▶ Shinder, D. (2011). Networking For Dummies (8th ed.). John Wiley & Sons.
- ▶ Stallings, W. (2013). Data and Computer Communications (10th ed.). Pearson.
- ▶ Peterson, L. L., & Davie, B. S. (2011). Computer Networks: A Systems Approach (5th ed.). Morgan Kaufmann.
- ▶ Postel, J. (1981). Internet Protocol DARPA Internet Program Protocol Specification (RFC 791). Internet Engineering Task Force.